

Lumina: Confidence-Weighted Identity Fusion and Preference-Aware Ranking for Photo Curation

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Abstract—Lumina curates a personal photo batch into events and people, then selects a best shot of each person at each event. The Semester 1 prototype already combined face embeddings, body re-identification, scene embeddings and a weighted aesthetic ranker, but identity was not fused, event boundaries depended on who appeared in the frame, and the ranker could not express a blink, a near-duplicate burst, or a user’s taste. This paper specifies the capstone model. Faces are matched to person boxes by a one-to-one assignment with a head-position prior. ArcFace and OSNet embeddings are placed in separate blocks of a concatenated vector, scaled by a confidence weight $\alpha \in [0.7, 1]$, so that cosine similarity in the fused space is an α -weighted blend of the two modalities. Events are agglomerative clusters on DINOv3 distances blended with capture time, named by CLIP against a closed prompt bank. Ranking uses seven signals with floored normalisation, an absolute eyes-open score, and a hard constraint that a detected blink loses to any eyes-open alternative. A Bradley–Terry update turns a user’s swap into one gradient step on the weight vector. On a private set of 55 photographs in 20 occasions, the label-free event threshold reaches an adjusted Rand index of 0.985 against the folder grouping, against 0.549 for the Semester 1 event features, and identity clustering makes none of the 526 merges that would put two faces from one photograph in the same person. Simulated users with known tastes, scored on the real per-event signals, raise held-out top-1 agreement by 0.03 to 0.26 after a handful of swaps. No labelled public album and no human study are reported.

Index Terms—photo curation, face recognition, person re-identification, event clustering, learning to rank, preference learning

I. Introduction

A holiday, a wedding or a family weekend produces a pile of near-duplicate photographs. The useful output is not another grid of files. It is a small set of events, a cast of people, and one defensible shot of each person in each event. That task has three coupled decisions: who is the same person when the face is small, turned or missing; which photographs belong to the same occasion when the background barely changes; and which frame is the one to keep when several are acceptable.

This paper continues the Semester 1 baseline for the same system. The baseline clustered identities with HDBSCAN, grouped events from scene embeddings plus a person co-attendance vector, and ranked photos with a fixed seven-signal score. The model changes in Sections III-B–III-E are the capstone contribution. Software structure is summarised only in Section IV-C.

Classical photo browsers answered the second question with time and colour [1]. Modern phone galleries add face clustering. Neither, on its own, ranks a burst, explains the choice, or changes its mind when the user prefers a different frame. Lumina is a batch pipeline for that joint decision. It does not train a new foundation model. It composes published encoders and puts the modelling effort into the fusion, the clustering objective, the constraints on the ranker, and a preference update that can move from a handful of pairwise edits.

The Semester 1 baseline detected people with YOLOv8 [5], embedded bodies with OSNet [4], clustered them with HDBSCAN [6], embedded scenes with a DINO-family model, scored aesthetics with NIMA [7], and ranked each event with seven hand-set weights. Its own measurements, on a few dozen photographs, already pointed at the next model rather than at more of the same pipeline. HDBSCAN’s silhouette on re-identification embeddings was about 0.25, against about 0.79 for a tuned DBSCAN. Adding a person co-attendance block to the event features raised silhouette only from 0.764 to 0.775. An ablation of the ranker found that the eye-aspect term never changed a winner, which is evidence that the term had no usable dynamic range, not evidence that blinks are unimportant.

This paper treats those findings as the problem statement. The contributions are:

- 1) a confidence-weighted concatenation of ArcFace [2] and OSNet that keeps cosine similarity equal to a modality blend, with a floor on the face weight so that a face with a body and the same face without one still meet;
- 2) event clustering on time-blended DINOv3 distances, dropping identity from the event features so that a clustering error about people cannot define the occasion;
- 3) a ranker whose normalisation, eyes-open score and blink constraint are specified so that burst noise and closed eyes cannot win on a rescaled aesthetic gap;
- 4) an online Bradley–Terry update [8] of the seven weights, with the step size chosen on simulated users and checked on a held-out draw of the same kind;
- 5) an identity check, using the same ArcFace model, that discards a generative retouch when the weakest

face match falls below a set cosine.

Items 1–4 are the core model and are what Section V measures. Naming, search, captions and retouching are supporting components (Section IV): they read the core model’s output and do not change who, which event, or which photograph. The paper does not claim a labelled accuracy on a public album benchmark.

II. Related work

PhotoTOC clustered personal photographs by creation time and colour histograms and showed that an automatic overview made search easier than folders alone [1]. The modelling choice that still matters is the use of time as an event feature, not only appearance. Later galleries replaced colour with learned descriptors and added face groups. Lumina follows that split — scene structure and identity are different clusterings — and adds an explicit ranker between them.

ArcFace trains a face embedding with an additive angular margin, so the natural comparison is cosine similarity on the hypersphere [2]. InsightFace’s buffalo_1 package is the detector and the 512-dimensional embedding used here [3]. Person re-identification covers the frames where the face is absent. OSNet’s omni-scale residual blocks produce a 512-dimensional body descriptor from a 256×128 crop [4]; YOLOv8-nano supplies the boxes [5]. The two embeddings are not in one linear space. Averaging them mixes incompatible coordinates. Late fusion of similarities, or an equivalent concatenation, is the construction that respects that fact. HDBSCAN remains appropriate when density varies and no threshold is known [6]; it is a poor default once the embedding was trained for a cosine threshold, which is the face case.

Self-supervised vision transformers in the DINO line provide a scene descriptor that is not tied to a closed label set [9], [10]. We use DINOv3 ViT-S/16, with DINOv2-base as a fallback if those weights are unavailable. CLIP maps images and text into one space and is already a zero-shot classifier when the label set is a list of prompts [11]. That is the right tool for naming an event and for text search over a few hundred vectors; an approximate nearest-neighbour index is unnecessary at this size. BLIP is used only to caption the selected frame [12]. It is not a feature in the clusterer or the ranker.

NIMA predicts a distribution over aesthetic ratings on AVA and reports the mean as a score from 0 to 10 [7], [13]. Aesthetic score is one term in the ranker, not the ranker. Maximal marginal relevance (MMR) trades that relevance against redundancy when the user wants a set rather than a single winner [14]. Bradley–Terry models a pairwise preference as a logistic of the difference in scores and is identified from comparisons without asking the user for a numeric rating [8]. Silhouette score is the internal criterion used to pick an agglomerative threshold [15]; it is not a substitute for a label.

III. Model

A job is a set of images $\{I_i\}_{i=1}^N$, N up to a few hundred. Each image may carry an EXIF capture time t_i . The model returns a partition of images into events, a set of identity clusters, a total order inside each event, a best photograph per person inside each event, three highlight sets, optional text-query ranks, and a weight vector that later jobs can reuse. Figure 1 is the data flow. Encoders are frozen. Everything after the embeddings is deterministic given the configuration in Table I.

We separate the core model from supporting components. The core model is the chain that decides the curation: face–body association and identity fusion (Sections III-A–III-B), event clustering (Section III-C), the best-shot ranker (Section III-D) and preference learning (Section III-E). Every experiment in Section V targets one of these four. Supporting components (Section IV) name events, answer text search, caption and retouch. They read the core model’s output and never feed back into who, which event, or which photograph.

A. Associating a face with a body

Semester 1 linked a face to the first person box that contained it. Overlapping people in a group photograph then shared a face. The replacement is a greedy one-to-one assignment. For face box f and person box p , let $c(f, p)$ be the fraction of the face area inside the person box. Pairs with $c(f, p) < 0.5$ are discarded. Among the rest the score is

$$c(f, p) + 0.5 h(f, p), \quad (1)$$

where the head prior h is high when the face centre is near the horizontal centre of the person box and in its upper portion. With person width w and height h_p , and face centre (x, y) ,

$$h = \max(0, 1 - \Delta x) \max(0, 1 - \min(\Delta y/0.6, 1)), \quad (2)$$

$\Delta x = |x - \bar{x}_p|/w$ and $\Delta y = (y - y_p^{\text{top}})/h_p$. Candidates are taken in descending score. Each face and each person box is used at most once. A body with no assigned face, and a face with no assigned body, both remain in the pipeline; fusion treats the missing block as empty rather than imputing it.

B. Confidence-weighted embedding fusion

Let $\hat{f}$ and $\hat{b}$ be L2-normalised ArcFace and OSNet vectors. They are not added. The fused descriptor is the concatenation

$$u = \begin{bmatrix} \sqrt{\alpha} \hat{f} \\ \sqrt{1 - \alpha} \hat{b} \end{bmatrix}, \quad (3)$$

which already has unit norm when both blocks are present. For two detections with weights α_i and α_j ,

$$u_i^T u_j = \sqrt{\alpha_i \alpha_j} \hat{f}_i^T \hat{f}_j + \sqrt{(1 - \alpha_i)(1 - \alpha_j)} \hat{b}_i^T \hat{b}_j. \quad (4)$$

When $\alpha_i = \alpha_j = \alpha$ this is exactly α times the face cosine plus $(1 - \alpha)$ times the body cosine. A missing modality

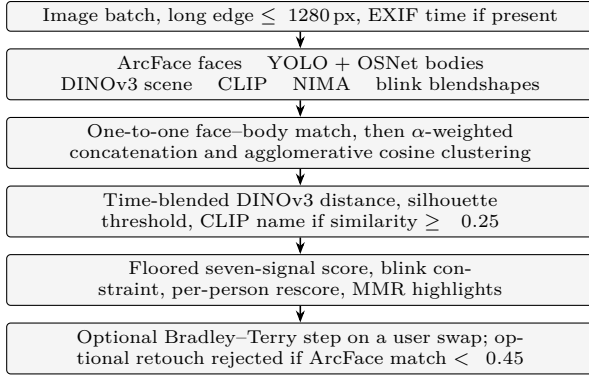


Fig. 1. Capstone inference path. Encoders are pretrained and frozen. Identity and events are separate partitions. Preference learning and retouching run only when the user asks.

TABLE I
Operating points used by the capstone model.

Choice	Value
Face embedding	InsightFace buffalo_1, 512-d
Body embedding	OSNet-x1.0, 256 × 128, 512-d
Scene embedding	DINOv3 ViT-S/16 (DINOv2-base fallback)
Text-image	CLIP ViT-B/32
Aesthetic	NIMA (AVA), one scalar in [0, 10]
Face cluster distance	0.60 cosine (face-only ablation)
Fused cluster distance	0.58 cosine
Body attach distance	0.75 Euclidean to ReID centroid
α range	[0.7, 1.0]
Event time weight, τ	0.35, 6 hours
Event threshold sweep	[0.05, 1.0] step 0.025, patience 10
Duplicate cosine	≥ 0.965 on DINOv3
Blink reject	eyes-open < 0.45 and a face was detected
MMR λ	0.90 / 0.70 / 0.45 (at most 6)
Preference step, prior pull	$\eta = 0.20$, $\lambda_{\text{prior}} = 0.05$
Retouch reject / warn	ArcFace cosine 0.45 / 0.65

forces α to 1 (face only) or 0 (body only) and zeros the other block, so similarity falls back to the modality the two detections share.

The weight itself is a detection-quality score, not a learned gate:

$$\gamma = \text{clip}\left(\frac{s_{\text{det}} - 0.35}{0.85 - 0.35}, 0, 1\right), \quad (5)$$

$$\sigma = \text{clip}(r_{\text{face}}/0.02, 0, 1), \quad (6)$$

$$\alpha = 0.7 + 0.3\gamma(0.5 + 0.5\sigma). \quad (7)$$

Here s_{det} is the InsightFace detection score and r_{face} is the face area as a fraction of the image. A large, confident face gets $\alpha = 1$. A weak or tiny face gets α near 0.7, never below. The floor is there because of (4). A face that was matched to a body at $\alpha = 0.7$ and the same face with no body ($\alpha = 1$) have face-term scale $\sqrt{0.7} \approx 0.837$. Take a face cosine of 0.50 as a worked example, not a measured average: scaled by $\sqrt{0.7}$ it becomes a fused similarity of about 0.42, i.e. a cosine distance of about 0.58, which sits on the fused threshold. The agglomerative threshold is

0.58 for that reason. The face-only ablation, which is the Semester 1 geometry, uses 0.60.

Clustering is average-linkage agglomerative clustering on cosine distance [16]. HDBSCAN on OSNet, with noise points reassigned to the nearest centroid, is used only when the batch has no usable face embedding, and as the body_only ablation. After face clusters exist, a person box that still has no identity is attached to the nearest ReID centroid if the Euclidean distance is below 0.75.

A final filter decides which clusters are subjects rather than background. Public scenes produce a singleton cluster for every bystander. A cluster is shown if it appears in at least two photographs or if its largest face covers at least 0.004 of some image (a square face of about 80 pixels on the 1280-pixel working resolution). Other clusters stay in the index and do not enter the cast.

C. Events from scene and time

Semester 1 concatenated a DINO vector with a multi-hot of person identities and multiplied the person block by a weight w_{person} chosen to maximise silhouette. On 16 images the best silhouette moved from 0.764 at $w = 1$ to 0.775 at $w = 5$, and the cluster count moved from 5 to 7. Two gatherings in the same house were the motivating case. The gain is small, and the feature has a structural fault: an identity error changes the event feature of every frame that person appears in, and the same household attending two occasions is exactly when co-attendance should not merge or split the day by itself.

The capstone event feature is the DINOv3 embedding only. Let rows of X be L2-normalised. Visual distance is $D_{ij}^{\text{vis}} = \max(0, 1 - x_i^T x_j)$. Where both photographs have EXIF time,

$$D_{ij} = (1 - w) D_{ij}^{\text{vis}} + w \min(|t_i - t_j|/\tau, 1), \quad (8)$$

with $w = 0.35$ and $\tau = 6$ hours. A missing timestamp leaves the pair at its visual distance, so a batch stripped of EXIF degrades to pure scene clustering rather than failing. Average-linkage agglomerative clustering is run on D across distance thresholds from 0.05 to 1.0 in steps of 0.025. The threshold with the best silhouette on the precomputed distance [15] is kept, and the sweep stops after ten consecutive thresholds that do not improve. The earlier sweep, from 0.1 to 3.0 in steps of 0.25 with patience 3, is the one Section V-C shows landing on the wrong side of the silhouette peak. Blended distances already lie near [0, 1], so extending the grid past 1 added nothing.

Equation (8) has a sharp operating consequence. Two visually identical photographs ($D^{\text{vis}} = 0$) taken more than τ apart have blended distance $w = 0.35$. They can be separated only if the selected threshold is below 0.35. A silhouette choice of 0.6, which the Semester 1 sweeps sometimes produced, will keep those two frames together. Time is a nudge of bounded size, not a hard cut. That is deliberate: a wrong clock should not override a clearly

different scene, because the visual term still carries weight 0.65.

D. The seven-signal ranker

Inside an event, each photograph receives a composite score $S = \sum_{k=1}^7 w_k z_k$ with $\sum_k w_k = 1$ and z_k a normalised signal. Table II lists the signals and the capstone weights. Two weights differ from Semester 1: eye openness moves from 0.05 to 0.10, and detection confidence from 0.10 to 0.05. Detection confidence measures whether a box is reliable. It is a weak statement about whether the photograph is the one to keep. Eye openness is the opposite, once it is measured on an absolute scale.

Plain min-max normalisation is the wrong map for a burst. If every NIMA score in the event lies between 5.00 and 5.05, min-max stretches that gap to 0 versus 1 and the aesthetic term dominates the rank for no visual reason. Floored min-max divides by $\max(\max z - \min z, c_k)$ with the floors in Table II. A 0.05-point NIMA gap then stays near $0.05/0.75 \approx 0.07$ instead of 1. Centrality uses the same idea on Euclidean distance to the mean embedding. The mean of unit vectors is not itself unit, so this is distance-to-centroid, not cosine-to-a-second normalisation. The Semester 1 write-up described centrality as cosine to the centroid; the implementation now matches the floored Euclidean form above.

Pose is

$$q = \text{clip}\left(1 - 0.5\frac{|\text{yaw}|}{90} - 0.3\frac{|\text{pitch}|}{90} - 0.2\frac{|\text{roll}|}{45}, 0, 1\right), \quad (9)$$

with angles in degrees. Geometric eye-aspect ratio (EAR) from the detector’s 68 landmarks [17] does not separate the cases we care about. An informal diagnostic recorded with the blink module, not a labelled test set, found closed and open eyes both near 0.20 (about 0.199 against 0.209) on the regularised landmarks. The ranker therefore prefers MediaPipe’s trained blink blendshapes [18]. Eyes-open is $1 - \max(\text{blink}_L, \text{blink}_R)$. If the landmarker misses the face, the fallback is $\text{clip}((\text{EAR} - 0.15)/0.10, 0, 1)$, which maps a blink to 0 and a clearly open eye to 1 without looking at the rest of the event. In a group photograph the photo-level eyes-open score is the minimum across faces, so one blinking person disqualifies the frame. Sharpness, pose and detection confidence are averaged; face size takes the maximum.

The score alone still lets a technically sharp, high-NIMA blink win, which is the failure that the weight change does not fix. After scoring, any photograph with a detected face and eyes-open below 0.45 is ordered after every non-blinking photograph. Ties inside each block break by S . If every frame is a blink, the constraint does not empty the event: the best blink remains the selection, and it is not flagged as a reject. The same rule is applied when each person’s own photographs are re-scored. For that per-person order, centrality and NIMA stay at their event-level values, and the five face signals are recomputed on that

TABLE II
Ranker weights. Semester 1 weights in parentheses where they differ. Centrality and the five face signals other than eyes use floored min-max inside the event. Eyes-open is absolute.

Signal	Weight	What is measured
Centrality	0.25	1 minus floored min-max of $\ x - \bar{x}\ _2$ to the event’s DINOv3 mean. Floor span 0.06.
NIMA	0.25	Aesthetic mean in $[0, 10]$. Floor span 0.75 points.
Sharpness	0.15	Laplacian variance of the face region. Floor span 60.
Face size	0.10	Face area over image area. Floor span 0.008.
Detection	0.05 (0.10)	InsightFace score. Floor span 0.10.
Pose	0.10	Frontal score from yaw, pitch and roll, Eq. (9). Floor span 0.25.
Eyes	0.10 (0.05)	Eyes-open in $[0, 1]$, not rescaled.

person’s faces. Eyes stay absolute. The other four face signals are ordinary min-max across that person’s frames in the event; the floors of Table II are not reapplied there. A very small sharpness gap can therefore still dominate a one-person burst. That inconsistency is recorded in Section VI.

Near-duplicates are the connected components of DINOv3 pairs with cosine at least 0.965. The highest-ranked member of a component is the keeper. The others are marked as duplicates of it. A non-winner is also flagged when the face Laplacian variance is below 25, or when eyes-open is below 0.45. Photographs with no face are not flagged for blur or blink. The selected photograph of the event is never flagged.

Explanations are templates over the normalised signals, not a language model. The summary names the strongest one or two terms, and if the gap to the runner-up on some term exceeds 0.25 it names that term as the margin. A lower-ranked photograph names its weakest term among those with weight at least 0.10, if that term is below 0.4. The text is a reading of the score, so it cannot justify a decision the score did not make.

MMR builds a set of at most six highlights,

$$i_{t+1} = \arg \max_{i \notin S_t} \left[\lambda S_i - (1 - \lambda) \max_{j \in S_t} \cos(x_i, x_j) \right], \quad (10)$$

with x the DINOv3 embedding and $\lambda \in \{0.90, 0.70, 0.45\}$ for a quality-leaning, balanced, or diverse set [14]. The first pick is $\arg \max_i S_i$. Because the blink constraint reorders the best-shot list without changing S_i , that first highlight can be a blink the best-shot rule rejected. The two outputs are not the same policy.

A separate MediaPipe Face Mesh pass (468 landmarks) can score symmetry and proportions for a display view. Those scores are not terms in S .

E. Preference learning

A swap — the user promotes photograph B over the system’s choice A — is one Bradley-Terry observation.

With normalised signal vectors s_A and s_B and weights w on the simplex,

$$P(B \succ A) = \sigma(w^\top (s_B - s_A)), \quad \sigma(u) = (1 + e^{-u})^{-1}. \quad (11)$$

One gradient step on the log-likelihood, with a quadratic pull toward the default w_0 of Table II, is

$$w \leftarrow \Pi(w + \eta [(1 - p)(s_B - s_A) - \lambda_{\text{prior}}(w - w_0)]), \quad (12)$$

where p is (11) under the current w , $\eta = 0.20$ and $\lambda_{\text{prior}} = 0.05$. The step size is the result of the grid in Section V-E, not a round number chosen in advance. The map Π clips each coordinate at 0.01 and renormalises. The prior term is the point of the design. A few swaps should move the ranker, not replace a tuned default with a one-hot vector fit to noise. The same clip is why a user who truly cares about a single signal cannot be represented exactly: every weight stays at least 0.01 before normalisation, and the pull keeps mass on w_0 .

Weights persist per user and can be applied to any later session. They do not retrain ArcFace, OSNet, DINO, CLIP or NIMA.

IV. Supporting components

A. Naming and search

Each event is named by comparing the L2-normalised centroid of its CLIP image embeddings with a bank of 26 prompts (“a photo taken at a wedding ceremony”, “a photo of people eating at a food court or hawker centre”, and so on). The winning prompt is accepted only if its cosine is at least 0.25. The floor was set above the band where unrelated ViT-B/32 pairs commonly score (about 0.20–0.23 in the implementation notes); we did not re-estimate that band. Below the floor the event stays unnamed by CLIP. A refused name falls back to a time phrase from EXIF (“Saturday afternoon”) and, if there is no time, to “Moment k ”. Duplicate titles are disambiguated with the time phrase.

Text search embeds the query with four templates (“a photo of {}”, “a photo of a {}”, “a photograph featuring {}”, and the raw string), averages the normalised vectors, and ranks images by cosine. At a few hundred photographs the scan is exact. An optional vision-language call may replace the short CLIP label with a two-to-five-word album title. The prompt forbids inventing a person’s name, a place or a date, and requires the scene tag to be one of the 26 labels or null. If the call fails, the CLIP or time label remains. BLIP-base writes one caption of the event’s selected photograph (beam search, 32 new tokens) and is discarded on any error. Neither the caption nor the album title is an input to clustering or ranking.

B. Identity-guarded retouching

An optional edit sends the selected photograph to a hosted image model with a prompt restricted to skin, eyes, exposure, colour and noise, and with an explicit ban on

reshaping features, changing age or ethnicity, reframing, or adding people. The edited image is not shown on the strength of that prompt. Every face in the original is re-embedded with the same ArcFace model and matched to its nearest face in the edit. The identity score is the minimum of those cosines, so one altered face in a group fails the check even if the others match. Below 0.45 the edit is discarded. From 0.45 to 0.65 it is shown with a warning. At or above 0.65 it is accepted. If no face is found, the edit is shown and marked unverified. These cutoffs are operating points in the region where buffalo_1 same-person pairs are expected to land; they were not fit on a labelled retouch corpus.

A second control is about the endpoint rather than the face. A text-to-image interface ignores the input photograph and can return people who were not in it. One observed call did exactly that. The client refuses models that are served only on that interface. The original file is never overwritten.

C. Where the model runs

The encoders and the mathematics above run in one Python service behind a web client. Jobs are processed one at a time because the loaded networks are not thread-safe. Features are cached by the SHA-256 of the file bytes, with a version salt that invalidates the cache when extraction changes. A repeated batch therefore skips inference. Progress is streamed to the client. Photographs leave the service only for the optional retouch and the optional album-title call. Account storage and the printable album are consumers of the ranking; they do not change it.

V. Evaluation

The core model makes four decisions, and each experiment below asks whether that decision is doing what it claims. Identity asks whether the same person is grouped together. Events ask whether photographs of one occasion stay together. The ranker asks which of its rules actually change the chosen photograph. Preference learning asks whether a few corrections move later choices toward a known taste. Naming, search, captions and retouching are not part of this measurement: none of them feeds back into those four decisions.

A. Dataset and what counts as correct

The set is 55 private photographs that the authors grouped into 20 occasions. The occasion label is the leading number of the file name, so 9-1 and 9-2 are the same occasion by construction. That grouping is one person’s judgement of where an occasion starts, which is why event quality is also reported with indices that do not use the labels. Nobody labelled who is who, so identity is not scored against a person list. Thirty-two of the 55 photographs carry an EXIF capture time.

All four experiments use the encoders the service uses, on CPU, with the DINOv3 ViT-S/16 weights actually

loaded rather than the DINOv2 fallback. Random draws use seed 42. A check that matters for the rest of the section: the event partition the harness computes is identical to the partition the pipeline returns. A cold pass over the 55 photographs takes 82s and a repeat from the feature cache takes 1.9s.

B. Identity, without person labels

Two faces in the same photograph cannot be the same person. Counting how often a clustering does that is a mistake count that needs no labelling, and it is the number we trust most. The second check is stability: drop a random 20% of the detections, cluster again, and measure the adjusted Rand index (ARI) [19] against the clustering of the same detections in the full run, over 30 draws. A clustering that reshuffles when a few photographs are removed is not a reliable cast list. Silhouette is reported only as a description of separation.

Table III is the comparison. Fusing the face and the body, and using the face alone, make none of these impossible merges. Clustering the body alone makes 85 of them, which is the empirical reason the face block carries most of the weight in (3). The production threshold of 0.58 sits inside a band that stays clean: raising it through 0.45, 0.50, 0.55, 0.58 and 0.62 keeps the cannot-link count at zero, and the first violations appear at 0.66 (3 of 526). Stability stays between 0.94 and 0.99 across that sweep, so the threshold is not balanced on one value. Lowering the face-weight floor from 0.7 to 0.5, or raising it to 0.9, also leaves the cannot-link count at zero. The floor is justified by the scaling argument in Section III-B, and this set does not contradict it. It also does not prove the floor; a set with more occluded faces might.

C. Events against the folder grouping

Event quality uses the folder labels, and we say so wherever a number depends on them. ARI is 1 when the clustering matches the grouping exactly and 0 when it matches no better than chance. Homogeneity is 1 when no cluster mixes two occasions; completeness is 1 when no occasion is split. The threshold itself is chosen by silhouette, which does not see the labels, and the same partition is then scored against them. An oracle row records the best ARI on the same distance matrix if the labels were allowed to pick the threshold. Stability is again an 80% subsample, here of photographs, over 30 draws.

Table IV is the result, and Figure 2 is why the sweep changed. The coarse grid of Section III-C’s predecessor, steps of 0.25 from 0.1, has its silhouette maximum at the first point it tries. That cuts the 20 occasions into 27 clusters (ARI 0.745). Steps of 0.025 put the silhouette peak at 0.175, which is also the threshold the oracle would have chosen, and the ARI rises to 0.985 with 21 clusters. The early stop does not cause it: patience 3, patience 10 and a full sweep of the fine grid all select 0.175 on this matrix. One occasion is still split. We do not have a label

TABLE III

Identity clustering on 55 photographs. Cannot-link counts pairs of faces in one photograph that were assigned the same person. Stability is the mean ARI over 30 subsamples of 80% of the detections.

Mode	Clusters	Shown	Cannot-link	Stability
Fused (production)	49	42	0 / 526	0.973
Face only	50	41	0 / 526	0.980
Body only	86	83	85 / 976	0.880

TABLE IV

Event clustering. The oracle row uses the folder labels to pick the threshold and is an upper bound, not a method. Semester 1 is scene features plus a person co-attendance block and no time.

Setting	Thr.	k	ARI	Stability
Production	0.175	21	0.985	0.969
Coarse sweep	0.100	27	0.745	0.749
No time, $w=0$	0.225	22	0.886	0.957
Semester 1 features	0.300	21	0.549	0.874
Oracle, this distance	0.175	21	0.985	—

for which one, and we do not claim the remaining error is the algorithm rather than the grouping.

Time is a smaller lever than the grid, but it is not zero. With the fine sweep and no time term the ARI is 0.886; any time weight from 0.2 to 0.5 reaches 0.985, and 0.7 falls back to 0.954. Changing τ from 1 hour to 24 hours, or switching between average, complete and single linkage, leaves the ARI at 0.985. The Semester 1 features, rescored with the same fine sweep, reach only 0.549, and two of their clusters mix photographs taken on different dates. The production clustering mixes none. Its best ARI at any threshold is 0.600, so the gap is in the features, not in the threshold rule.

D. What moves the best shot

There is no label for the best photograph of an occasion, and this paper does not invent one. The question the set can answer is which part of the ranker changes the choice. Sixteen of the 20 occasions contain at least two photographs. For each variant we count how many of those 16 change their winner relative to the production ranker, the mean Kendall τ between the two orders, and how often the winner is a detected blink when the same occasion also contains an eyes-open frame. NIMA scores on this set run from 3.26 to 4.88, so the aesthetic term has real range here; a bug that stored every score as zero was found and cleared before these numbers were taken.

Table V separates three claims. The blink constraint is not cosmetic: without it, 3 of the 16 winners are closed-eye frames that had an open-eye alternative, and the Semester 1 rule, which also lacks the floors, does the same. An aesthetic score on its own is not a curator. It disagrees with the full ranker on 12 of 16 winners and agrees with the production order worse than a reversed ranking. Centrality and sharpness each move 4 winners;

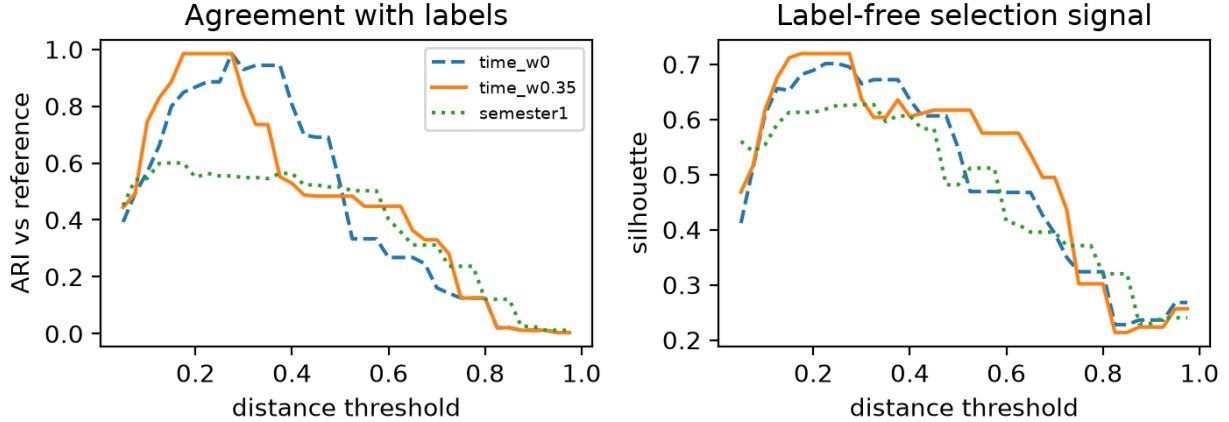


Fig. 2. Event distance threshold on the 55 photographs. Left: agreement with the folder grouping. Right: the silhouette the selection rule actually sees. The production rule takes the silhouette peak of the solid curve.

TABLE V

Best-shot ranker on 16 occasions with at least two photographs. Blink picks are winners with eyes-open below 0.45 in an occasion that had an eyes-open alternative.

Variant	Changed	Kendall τ	Blinks
No span floors	4 / 16	0.66	0
No blink constraint	3 / 16	0.70	3
Semester 1 rule	8 / 16	0.32	3
NIMA alone	12 / 16	-0.15	4
Without centrality	4 / 16	0.64	0
Without sharpness	4 / 16	0.68	0
Without NIMA	1 / 16	0.94	0

removing NIMA moves 1, and removing face size, detection confidence, pose or the eyes-open weight moves none. The weight can be inert while the constraint is not, which is the same reading as the Semester 1 ablation, now with the constraint present to take over the job. Perturbing any single weight by $\pm 20\%$ changes no winner. The ranking is stable to small jitter on this set, and that is all the perturbation says.

E. Preference learning on held-out occasions

A swap is only evidence of learning if it changes a choice the model was not trained on. The signals are the real normalised vectors from the 16 occasions. The users are simulated, because their taste has to be known. Three are specified by hand: an aesthete who puts weight 0.60 on NIMA, a portraitist who puts 0.35 on sharpness and 0.25 on face size, and a storyteller who puts 0.55 on centrality. Sixty more draw a weight vector from a symmetric Dirichlet with concentration 0.7, which favours users who care about a few signals. Each occasion is scored by that hidden vector.

One split holds out 30% of the occasions. On the rest, the simulated user is shown the system’s current pick and, where they disagree, the disagreement is one update (12). After each pass over the training occasions the learned

weights pick a photograph on the held-out occasions, and we record whether it is the one the hidden vector would pick. Chance, from the number of photographs per occasion, is about 0.35. A second condition replaces the user’s choice with a softmax at temperature 0.05, so near-ties are sometimes swapped the other way. That is inconsistent judgement, not a 5% chance of a random click. The step size and the prior pull were chosen on a separate draw of 20 Dirichlet users plus the three personas, over a grid of seven step sizes and five pulls, and the winner was then scored on the fresh draw reported here. A candidate was eligible only if 60 synthetic swaps from a user who always prefers the higher NIMA score leave NIMA as the dominant weight, the same condition the unit tests enforce.

The grid selects $\eta = 0.20$ with the prior pull left at 0.05. Its mean gain across the eight tuning cells is $+0.131$, against $+0.117$ for the previous step of 0.35, and its worst cell loses less (-0.060 against -0.095). Table VI is the held-out confirmation, and Figure 3 shows the path. Where the untrained ranker is already far from the user, learning moves it: the aesthete goes from 0.51 to 0.78, and the regret of that gap, as a fraction of the best-to-worst score range inside the occasion, falls from 0.28 to 0.05. The 60 random users go from 0.63 to 0.80, against 0.35 for picking uniformly. The previous step size does not do this uniformly. On the same splits it drops the storyteller from 0.81 to 0.79 with consistent swaps and from 0.84 to 0.78 with noisy ones.

The limit is in the same table. With noisy swaps the storyteller falls from 0.84 to 0.82, and the portraitist’s gain is not distinguishable from zero ($p = 0.31$). No point on the grid improved every cell at once. Learning helps the users the default ranker serves badly, and it does not reliably help the ones it already serves well once their corrections are inconsistent.

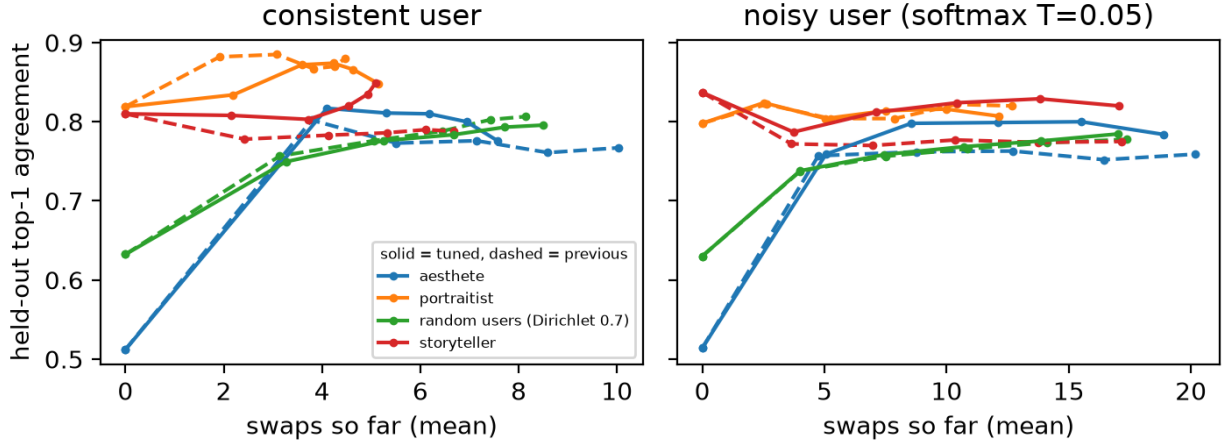


Fig. 3. Held-out agreement against the number of swaps. Solid lines use $\eta=0.20$; dashed lines use the previous step of 0.35.

TABLE VI
Held-out top-1 agreement for the selected step $\eta=0.20$, mean over 200 splits (40 per Dirichlet user). The p value is a one-sided Wilcoxon test of learned against untrained.

User	Noise	Untrained	Learned	p
Aesthete	none	0.512	0.776	<0.001
Portraitist	none	0.819	0.848	<0.001
Storyteller	none	0.810	0.849	0.028
60 random users	none	0.632	0.796	<0.001
Aesthete	$T=0.05$	0.514	0.784	<0.001
Portraitist	$T=0.05$	0.798	0.807	0.31
Storyteller	$T=0.05$	0.837	0.820	0.95
60 random users	$T=0.05$	0.630	0.785	<0.001

F. What these numbers do not show

The occasion labels are one grouping of one private collection, and the identity numbers say nothing about whether two photographs of the same person were correctly joined, only that the clustering avoids a particular impossible merge and does not reshuffle. The simulated users can only prefer mixtures of the seven signals the ranker already has, so the preference result does not show that a real user would accept the learned pick. A small study in which people choose the best frame, and then correct a few occasions before judging held-out ones, is the measurement that would. Recall of the text search was not measured.

VI. Discussion

The model is a set of constraints around frozen encoders. That is a good fit to the evidence we have, and a poor fit to a claim of general accuracy. Fusion assumes that a high α means the face embedding is the one to trust. ArcFace can be confidently wrong on a profile or a very small face; the size term in (7) only partly covers that, and there is no uncertainty model beyond α . The 0.58 threshold is justified by the floor calculation in Section III-B. Section V-B only shows that, on this collection, it sits

inside a band with no cannot-link violation. There are no identity labels, so the same number says nothing about people wrongly split apart. If same-person face cosines on other photographs sit lower than the 0.5 used in that calculation, the floor will split people; if different-person cosines sit higher, it will merge them.

On this collection the silhouette peak, the early-stopped sweep and the label-using oracle all select the same event threshold (Section V-C). That is one distance matrix. A later mode of silhouette on another batch can still be missed once patience runs out, and the folder labels are one person's grouping of 20 occasions. Time cannot separate visually identical frames once the threshold exceeds the time weight 0.35. Twenty-three of the 55 photographs have no EXIF time, so those pairs contribute none. Scene embeddings will merge two dinners in the same restaurant and will split one dinner if the lights change, which is the behaviour DINOv3 was trained to represent, not the behaviour of a calendar.

The ranker encodes one family's notion of a keeper: frontal, sharp, eyes open, reasonably large, aesthetically typical of the event. A documentary frame, a profile that is the only picture of someone, or a dark but important moment loses by construction. Preference learning can move the seven weights and cannot add a new notion of importance. Table VI is the quantitative form of that limit: the largest gains go to the user the default ranker serves worst, and one noisy user is slightly worse off after learning. Two further mismatches inside the ranker should not be smoothed over. Per-person face signals, other than eyes, are min-maxed without the floors that the event-level score uses. MMR reads the raw composite, so its first highlight can be a blink. Both are straightforward to align with the event-level policy; they are not aligned in the system this paper describes.

The retouch guard uses a single cosine cutoff and the weakest face. It does not detect a changed background,

a changed body, or a face the detector missed, and an unverified no-face edit is still displayed. Prompt text is not a guarantee. The guard is a filter on one observed failure mode.

What would change the strength of these claims is a public album with identity labels, a second person’s grouping of the occasions, Recall@ K for the prompt-ensembled CLIP queries, and a small study in which the swaps come from people rather than from a hidden weight vector. Until then the supported statement is the one in Section V: on this collection the fused clustering makes no cannot-link merge, the label-free event threshold matches the best labelled threshold, the blink constraint is what keeps closed eyes out of the winners, and a Bradley–Terry step of 0.20 improves held-out agreement for the users the default ranker fits least well.

VII. Conclusion

Lumina’s capstone model is the layer between frozen vision encoders and a photo selection. Identity is a cosine clustering of a confidence-weighted concatenation of face and body embeddings, after a one-to-one face–body assignment. Events are a time-blended clustering of scene embeddings, named only when CLIP clears a similarity floor. The ranker keeps the seven Semester 1 signals, changes how two of them are measured, refuses to let min–max invent contrast inside a burst, and refuses to let a detected blink outrank an eyes-open frame. A Bradley–Terry step adapts the weights from swaps. On simulated users scored with the real signals, a step of 0.20 raises held-out top-1 agreement by up to 0.26, and it does not do so for every user once the swaps are noisy. The same face embedding that builds identity also rejects a retouch that fails a minimum match. These are the claims the implementation and the measurements support. Identity labels and a human preference study remain open.

Acknowledgment

Semester 1 roles were split as follows: Ong Qing Quan integrated the baseline pipeline, the web application and deployment; Agustinus Benyamin Prasetyo designed the baseline notebook experiments and the baseline report. The capstone model in this paper is joint work on the shared system. An AI coding assistant was used to implement parts of the software and to draft this paper from the codebase, the Semester 1 report and the experiment

logs. The numbers in Section V come from one run of that harness, archived with the paper. The authors remain responsible for the text.

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